

Liam Haynie

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EDUCATION

Bachelor of Science in Mechanical Engineering, Emphasis in Robotics & Control

University of Utah

May 2027

GPA: 3.49

- **Dean's List: Spring 2025 – Fall 2025, Summer 2026**

Associate of Science in Biology

Utah Valley University

2019 - 2021

TECHNICAL SKILLS

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|-----------------------------------|--|--------------------------------|
| • SolidWorks Certified, CSWA 2025 | • Digital multimeter (DMM) | • 3D printing (FDM) |
| • MATLAB | • Soldering (through hole & SMD) | • Statistical data analysis |
| • Arduino (C/C++) | • Hands-on prototyping experience | • Experimental data collection |
| • LTSpice | • Slicing software (Cura & Bambu Studio) | • Adobe Premiere |
| • Oscilloscope operation | | • Microsoft Office |
| | | • Documentation management |

EXPERIENCE

R&D Engineering Intern, *Medtronic Neurovascular*

Jul. 2026 – Sep. 2026

- Characterized unreleased neurovascular guidewires to establish production warning limits and investigate product performance issues, supporting R&D testing and quality decisions.
- Designed and prototyped a universal fixture for microscopic inspection of intravenous neurovascular devices, improving consistency and efficiency of device inspection.
- Built an automated data-processing system for lot-monitoring tests, streamlining data organization and reducing manual processing.

Solution Specialist (Infectious Disease Antigen Testing Lab), *ARUP Laboratories*

Oct. 2020 – Jul. 2026

- Collaborated with LabTech and ARUP R&D teams on the development of a *Streptococcus pneumoniae* ELISA by performing assay testing and supporting experimental validation.
- Supported research and development of a rapid *Escherichia coli* O157:H7 card assay by sourcing, de-identifying, and preparing previously tested clinical samples for Meridian Bioscience's Curian device.

Undergraduate Research Assistant (UROP), *Additive Manufacturing Lab, University of Utah*

Jan. 2022 – Aug. 2022

- Conducted sponsored undergraduate research in additive manufacturing focused on the development of bioabsorbable sensors for periodontal aligners and artificial joint applications.
- Supported experimental testing by capturing, organizing, and analyzing data from laboratory experiments to evaluate sensor performance.
- Presented experimental methods and findings through written and oral research updates, contributing to project documentation and technical communication.

PERSONAL PROJECT

Rat Operated Car | Personal Project (inspired by [Crawford et al., 2020](#))

Jun. 2025 – Dec. 2025

- Designed and built an original rat operated vehicle for emotional enrichment using laser cut and 3D printed components, featuring a rat centric ergonomic carriage and a three-pad copper touch interface with Arduino control for rat directed steering.
- Programmed Arduino firmware with touchpad debouncing, a state tracking feedback system, and ramped PWM acceleration/braking to smooth motion and reduce jitter.